



Continuum Mechanics

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Chapter 04

Kinematics

Part 2



Cauchy-Green Strain



Definition (Principal Stretches and Directions)

Let $\phi : B \rightarrow B'$ be an admissible deformation, and let

$$F = \nabla \phi = RU = VR$$

be the right and left polar decompositions of the deformation gradient, where

$$U = \sqrt{F^T F}, \quad V = \sqrt{F F^T}$$

are symmetric positive definite and R is a rotation tensor.

Let $\{\lambda_i, \mathbf{u}_i\}$ and $\{\lambda_i, \mathbf{v}_i\}$ be eigensystems of U and V , respectively, ordered so that

$$\lambda_1 \geq \lambda_2 \geq \lambda_3 > 0.$$

The λ_i are the *principal stretches* associated with F . The $\mathbf{u}_i$ and $\mathbf{v}_i$ are, respectively, the *right* and *left principal directions*.



Definition (Right Cauchy–Green Strain Tensor)

Consider a general admissible deformation

$$\phi : B \longrightarrow B'$$

with associated deformation gradient $F = \nabla \phi$. The tensor-valued field $C : B \rightarrow \mathcal{V}^2$ defined by

$$C = F^T F$$

is called the *right Cauchy–Green strain tensor field* associated with ϕ .

At every point $\mathbf{X} \in B$, the tensor $C(\mathbf{X})$ is symmetric positive definite.



Why the Right Cauchy–Green Tensor Measures Stretch

Using the right polar decomposition $F = RU$, with

$$R^T R = I,$$

we obtain

$$\begin{aligned} C &= F^T F \\ &= (RU)^T (RU) \\ &= U^T R^T R U \\ &= U^2. \end{aligned}$$

Thus C contains the stretch information in F but not its rotational part.

In particular, the eigenpairs of C are

$$\{\lambda_i^2, \mathbf{u}_i\}, \quad i = 1, 2, 3.$$

This fact makes C especially useful as a measure of strain.



Proposition (Spectral Representation of an SPD Tensor)

Suppose $U \in \mathcal{V}^2$ is symmetric positive definite. Its eigenvalues are real and positive and may be ordered as

$$\lambda_1 \geq \lambda_2 \geq \lambda_3 > 0.$$

Furthermore, there is an orthonormal eigenbasis $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ such that

$$U\mathbf{u}_i = \lambda_i\mathbf{u}_i \quad (\text{no sum on } i), \quad \mathbf{u}_i \cdot \mathbf{u}_j = \delta_{ij},$$

and

$$U = \sum_{i=1}^3 \lambda_i \mathbf{u}_i \otimes \mathbf{u}_i, \quad U^2 = \sum_{i=1}^3 \lambda_i^2 \mathbf{u}_i \otimes \mathbf{u}_i.$$

Proof.

Exercise.





Result 4.4 (Local Stretch and Angle Change)

Let $\mathbf{X} \in B$ be arbitrary, let Ω be a ball of radius α centered at $\mathbf{X}$, and choose any unit vectors $\mathbf{e}, \mathbf{d} \in \mathcal{N}$, where

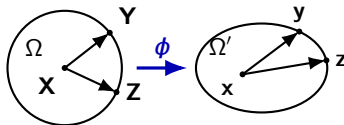
$$\mathcal{N} := \{\mathbf{v} \in \mathcal{V} \mid |\mathbf{v}| = 1\}.$$

Set

$$\mathbf{Y} = \mathbf{X} + \alpha \mathbf{e}, \quad \mathbf{Z} = \mathbf{X} + \alpha \mathbf{d},$$

so that $|\mathbf{Y} - \mathbf{X}| = |\mathbf{Z} - \mathbf{X}| = \alpha$. Under an admissible deformation $\phi : B \rightarrow B'$, write

$$\mathbf{x} = \phi(\mathbf{X}), \quad \mathbf{y} = \phi(\mathbf{Y}), \quad \mathbf{z} = \phi(\mathbf{Z}).$$





Result 4.4 (Cont.)

Let $C = F^T F$ be the right Cauchy–Green tensor at $\mathbf{X}$. For arbitrary $\mathbf{e}, \mathbf{d} \in \mathcal{N}$, define

$$\lambda(\mathbf{e}) := \sqrt{\mathbf{e} \cdot (C\mathbf{e})}.$$

Also define $\theta(\mathbf{e}, \mathbf{d}) \in [0, \pi]$ by

$$\cos(\theta(\mathbf{e}, \mathbf{d})) = \frac{\mathbf{e} \cdot (C\mathbf{d})}{\sqrt{\mathbf{e} \cdot (C\mathbf{e})} \sqrt{\mathbf{d} \cdot (C\mathbf{d})}}.$$



Result 4.4 (Cont.)

Then

$$\frac{|\mathbf{y} - \mathbf{x}|}{|\mathbf{Y} - \mathbf{X}|} \longrightarrow \lambda(\mathbf{e}) \quad \text{as } \alpha \downarrow 0. \quad (4.5_1)$$

Similarly,

$$\frac{|\mathbf{z} - \mathbf{x}|}{|\mathbf{Z} - \mathbf{X}|} \longrightarrow \lambda(\mathbf{d}) \quad \text{as } \alpha \downarrow 0. \quad (4.5_2)$$

If $s \in [0, \pi]$ is the angle between $\mathbf{y} - \mathbf{x}$ and $\mathbf{z} - \mathbf{x}$, then

$$\cos(s) \longrightarrow \cos(\theta(\mathbf{e}, \mathbf{d})) \quad \text{as } \alpha \downarrow 0. \quad (4.6)$$

Proof.

See the textbook, pages 122–123.





Remark (Directional Stretch)

The quantity $\lambda(\mathbf{e})$ is the limiting value of

$$\frac{|\phi(\mathbf{Y}) - \phi(\mathbf{X})|}{|\mathbf{Y} - \mathbf{X}|} = \frac{\text{deformed length}}{\text{initial length}}$$

as $\mathbf{Y}$ tends to $\mathbf{X}$ along the direction $\mathbf{e}$.

For this reason, $\lambda(\mathbf{e})$ is called the stretch in the direction $\mathbf{e}$ at $\mathbf{X}$.

The stretch is a local quantity: both C and $\lambda(\mathbf{e})$ generally depend on the material point $\mathbf{X}$.



Proposition (Stretch in a Principal Direction)

Let $\mathbf{u}_i$ be a right principal direction at $\mathbf{X}$ corresponding to the principal stretch λ_i , so that

$$U\mathbf{u}_i = \lambda_i\mathbf{u}_i \quad (\text{no sum on } i).$$

Then

$$\lambda(\mathbf{u}_i) = \lambda_i.$$

Proof.

Because $C = U^2$,

$$C\mathbf{u}_i = \lambda_i^2\mathbf{u}_i \quad (\text{no sum on } i).$$

Therefore

$$\begin{aligned}\lambda(\mathbf{u}_i) &= \sqrt{\mathbf{u}_i \cdot (C\mathbf{u}_i)} \\ &= \sqrt{\mathbf{u}_i \cdot (\lambda_i^2\mathbf{u}_i)} \\ &= \lambda_i,\end{aligned}$$

since $|\mathbf{u}_i| = 1$ and $\lambda_i > 0$.





Proposition (Maximum Directional Stretch)

Let $\phi : B \rightarrow B'$ be an admissible deformation, and suppose its principal stretches at $\mathbf{X}$ are ordered so that

$$\lambda_1 \geq \lambda_2 \geq \lambda_3 > 0.$$

Then

$$\max_{\mathbf{e} \in \mathcal{N}} \lambda(\mathbf{e}) = \lambda_1 = \lambda(\mathbf{u}_1),$$

where $\mathbf{u}_1$ is a right principal direction associated with λ_1 .

Proof.

Let $\mathbf{e} \in \mathcal{N}$ be arbitrary. In the orthonormal principal basis,

$$\mathbf{e} = \sum_{i=1}^3 c_i \mathbf{u}_i, \quad \sum_{i=1}^3 c_i^2 = 1.$$



Proof (Cont.)

Since $C\mathbf{u}_i = \lambda_i^2 \mathbf{u}_i$ with no sum on i ,

$$\begin{aligned}\lambda^2(\mathbf{e}) &= \mathbf{e} \cdot (C\mathbf{e}) \\ &= \left(\sum_{j=1}^3 c_j \mathbf{u}_j \right) \cdot C \left(\sum_{i=1}^3 c_i \mathbf{u}_i \right) \\ &= \left(\sum_{j=1}^3 c_j \mathbf{u}_j \right) \cdot \left(\sum_{i=1}^3 \lambda_i^2 c_i \mathbf{u}_i \right) \\ &= \sum_{j=1}^3 \sum_{i=1}^3 c_j \lambda_i^2 c_i (\mathbf{u}_j \cdot \mathbf{u}_i) \\ &= \sum_{i=1}^3 \lambda_i^2 c_i^2.\end{aligned}$$

Here i and j are dummy indices.



Proof (Cont.)

Because $\lambda_i \leq \lambda_1$ for $i = 1, 2, 3$,

$$\begin{aligned}\lambda^2(\mathbf{e}) &= \sum_{i=1}^3 \lambda_i^2 c_i^2 \\ &\leq \sum_{i=1}^3 \lambda_1^2 c_i^2 \\ &= \lambda_1^2 \sum_{i=1}^3 c_i^2 \\ &= \lambda_1^2.\end{aligned}$$



Proof (Cont.)

Since the stretches are positive,

$$\lambda(\mathbf{e}) \leq \lambda_1 \quad \text{for every } \mathbf{e} \in \mathcal{N}.$$

But $\lambda(\mathbf{u}_1) = \lambda_1$. Therefore

$$\max_{\mathbf{e} \in \mathcal{N}} \lambda(\mathbf{e}) = \lambda_1.$$



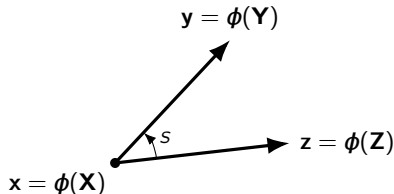
Local Change of Angle



Let $\Theta(\mathbf{e}, \mathbf{d}) \in [0, \pi]$ be the initial angle between the unit directions $\mathbf{e}$ and $\mathbf{d}$.
Let s be the angle between the deformed segments $\mathbf{y} - \mathbf{x}$ and $\mathbf{z} - \mathbf{x}$.

As $\alpha \downarrow 0$, the points $\mathbf{Y}$ and $\mathbf{Z}$ tend to $\mathbf{X}$ along $\mathbf{e}$ and $\mathbf{d}$, and

$$s \longrightarrow \theta(\mathbf{e}, \mathbf{d}).$$





Definition (Shear Between Two Directions)

The quantity

$$\gamma(\mathbf{e}, \mathbf{d}) := \Theta(\mathbf{e}, \mathbf{d}) - \theta(\mathbf{e}, \mathbf{d})$$

is called the *shear between the directions $\mathbf{e}$ and $\mathbf{d}$ at $\mathbf{X}$* .

Proposition (No Shear Between Orthogonal Principal Directions)

The shear between any two distinct principal directions, chosen as orthonormal eigenvectors of U , is zero:

$$\gamma(\mathbf{u}_i, \mathbf{u}_j) = 0, \quad i \neq j.$$

Proof.

Exercises 18 and 19.





Result 4.5 (Components of the Right Cauchy–Green Tensor)

Let C_{ij} be the components of C in an orthonormal frame $\mathcal{F} = \{\mathbf{e}_i\}_{i=1}^3$, so that

$$C_{ij} = \mathbf{e}_i \cdot (C\mathbf{e}_j).$$

Then, for any point $\mathbf{X} \in B$,

$$C_{ii} = \lambda^2(\mathbf{e}_i) \quad (\text{no sum on } i),$$

and, for $i \neq j$,

$$C_{ij} = \lambda(\mathbf{e}_i)\lambda(\mathbf{e}_j)\sin(\gamma(\mathbf{e}_i, \mathbf{e}_j)) \quad (\text{no sum}).$$



Proof.

By definition,

$$\cos(\theta(\mathbf{e}_i, \mathbf{e}_j)) = \frac{\mathbf{e}_i \cdot (C\mathbf{e}_j)}{\sqrt{\mathbf{e}_i \cdot (C\mathbf{e}_i)}\sqrt{\mathbf{e}_j \cdot (C\mathbf{e}_j)}}.$$

Therefore

$$C_{ij} = \lambda(\mathbf{e}_i)\lambda(\mathbf{e}_j)\cos(\theta(\mathbf{e}_i, \mathbf{e}_j)) \quad (\text{no sum}).$$

For $i \neq j$, orthonormality gives $\Theta(\mathbf{e}_i, \mathbf{e}_j) = \pi/2$, so

$$\gamma(\mathbf{e}_i, \mathbf{e}_j) = \frac{\pi}{2} - \theta(\mathbf{e}_i, \mathbf{e}_j).$$

Hence

$$C_{ij} = \lambda(\mathbf{e}_i)\lambda(\mathbf{e}_j)\sin(\gamma(\mathbf{e}_i, \mathbf{e}_j)).$$



Proof (Cont.)

For $i = j$,

$$\begin{aligned} C_{ii} &= \mathbf{e}_i \cdot (C\mathbf{e}_i) \\ &= \lambda^2(\mathbf{e}_i). \end{aligned}$$

Thus the diagonal components measure squared directional stretches, while the off-diagonal components measure the shear between orthogonal frame directions. □



Example (Simple Shear)

For the planar deformation $\phi(X_1, X_2) = (X_1 + X_2, X_2)$,

$$F = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad C = F^T F = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

Along the coordinate directions,

$$\lambda(e_1) = 1, \quad \lambda(e_2) = \sqrt{2},$$

and

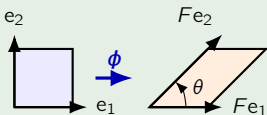
$$\cos(\theta(e_1, e_2)) = \frac{C_{12}}{\sqrt{C_{11}C_{22}}} = \frac{1}{\sqrt{2}}.$$

Hence $\theta = \pi/4$ and $\gamma = \pi/4$.

The eigenvalues of C and the corresponding principal stretches are

$$\mu_{\pm} = \frac{3 \pm \sqrt{5}}{2}, \quad \lambda_1 = \frac{1 + \sqrt{5}}{2}, \quad \lambda_2 = \frac{\sqrt{5} - 1}{2}.$$

Also, $\lambda_1 \lambda_2 = 1 = \det F$, so the shear preserves area.





Rigid Deformations



Definition (Rigid Deformation)

A homogeneous deformation $\phi : B \rightarrow B'$ is called *rigid* if

$$\phi(\mathbf{X}) = \mathbf{c} + Q\mathbf{X},$$

where $\mathbf{c} \in \mathcal{V}$ and $Q \in \mathcal{V}^2$ is a rotation tensor:

$$QQ^T = I, \quad \det(Q) = 1.$$

A rigid deformation is the composition of a rotation and a translation. Its deformation gradient is the constant tensor

$$F = Q.$$



Proposition (Rigid Deformations Produce No Strain)

For a rigid deformation, the right Cauchy–Green strain tensor is the identity:

$$C(\mathbf{X}) = I \quad \text{for every } \mathbf{X} \in B.$$

Thus a rigid deformation produces no strain.

Proof.

For the homogeneous deformation $\phi(\mathbf{X}) = \mathbf{c} + Q\mathbf{X}$,

$$F(\mathbf{X}) = Q.$$

Consequently,

$$C(\mathbf{X}) = F^T(\mathbf{X})F(\mathbf{X}) = Q^T Q = I \quad \text{for every } \mathbf{X} \in B.$$





Remark (Geometry of a Rigid Deformation)

For a rigid deformation, the relative position and orientation of any two points in B are unchanged by ϕ .

Indeed, for any $\mathbf{X}, \mathbf{Y} \in B$,

$$\begin{aligned}\phi(\mathbf{Y}) - \phi(\mathbf{X}) &= Q(\mathbf{Y} - \mathbf{X}), \\ |\phi(\mathbf{Y}) - \phi(\mathbf{X})| &= |Q(\mathbf{Y} - \mathbf{X})| \\ &= |\mathbf{Y} - \mathbf{X}|.\end{aligned}$$

Also, rotation tensors preserve angles. Thus a rigid deformation changes neither lengths nor angles.



Infinitesimal Strain



Definition (Infinitesimal Strain Tensor)

Consider an admissible deformation $\phi : B \rightarrow B'$ with associated displacement field $\mathbf{u} : B \rightarrow \mathcal{V}$,

$$\mathbf{u}(\mathbf{X}) = \phi(\mathbf{X}) - \mathbf{X}.$$

Define $E : B \rightarrow \mathcal{V}^2$ by

$$E = \text{sym}(\nabla \mathbf{u}) = \frac{1}{2} \left(\nabla \mathbf{u} + (\nabla \mathbf{u})^T \right).$$

The tensor E is called the *infinitesimal strain tensor field* associated with ϕ .



Proposition (Relation Between Finite and Infinitesimal Strain)

Let C be the right Cauchy-Green tensor associated with ϕ . Then

$$E = \frac{1}{2}(C - I) = \frac{1}{2}(\nabla \mathbf{u})^T \nabla \mathbf{u}.$$

Proof.

Because $\mathbf{u}(\mathbf{X}) = \phi(\mathbf{X}) - \mathbf{X}$,

$$\nabla \mathbf{u} = \nabla \phi - I = F - I.$$

Therefore

$$\begin{aligned} E &= \text{sym}(F - I) \\ &= \frac{1}{2}(F - I) + \frac{1}{2}(F^T - I). \end{aligned}$$



Proof (Cont.)

On the other hand,

$$\begin{aligned}\frac{1}{2}(C - I) - \frac{1}{2}(\nabla \mathbf{u})^T \nabla \mathbf{u} &= \frac{1}{2}(F^T F - I) - \frac{1}{2}(F^T - I)(F - I) \\ &= \frac{1}{2}F^T F - \frac{1}{2}I \\ &\quad - \frac{1}{2}(F^T F - F^T - F + I) \\ &= \frac{1}{2}F + \frac{1}{2}F^T - I \\ &= \frac{1}{2}(F - I) + \frac{1}{2}(F^T - I) \\ &= \text{sym}(F - I) \\ &= E.\end{aligned}$$





The Small-Displacement Approximation

The infinitesimal strain tensor is useful when

$$|\nabla \mathbf{u}| \ll 1.$$

Suppose, more precisely, that

$$|\nabla \mathbf{u}| = O(\epsilon).$$

Then

$$(\nabla \mathbf{u})^T \nabla \mathbf{u} = O(\epsilon^2),$$

and the preceding proposition gives

$$E = \frac{1}{2}(C - I) + O(\epsilon^2).$$

Thus, when displacements are small,

$$E \approx \frac{1}{2}(C - I).$$

This makes E a convenient measure of strain in the infinitesimal theory.



Result 4.6 (Stretch and Shear for Small Displacements)

Let E_{ij} be the components of E in an orthonormal frame $\mathcal{F} = \{\mathbf{e}_i\}_{i=1}^3$. If terms of order $O(\varepsilon^2)$ are sufficiently small to be neglected, then

$$E_{ii} \approx \lambda(\mathbf{e}_i) - 1 \quad (\text{no sum on } i),$$

and, for $i \neq j$,

$$E_{ij} \approx \frac{1}{2} \sin(\gamma(\mathbf{e}_i, \mathbf{e}_j)).$$

Here $\lambda(\mathbf{e}_i)$ is the stretch in the direction $\mathbf{e}_i$, and $\gamma(\mathbf{e}_i, \mathbf{e}_j)$ is the shear between $\mathbf{e}_i$ and $\mathbf{e}_j$.

Proof.

See the textbook, pages 126–127.





Remark (Reading the Components of E)

Result 4.6 gives a direct geometric meaning to the components of the infinitesimal strain tensor:

- E_{ii} approximates the change in length per unit initial length in the direction $\mathbf{e}_i$;
- $2E_{ij}$ approximates the sine of the shear between the initially orthogonal directions $\mathbf{e}_i$ and $\mathbf{e}_j$.

For small shear angles, $\sin \gamma \approx \gamma$, so

$$2E_{ij} \approx \gamma(\mathbf{e}_i, \mathbf{e}_j), \quad i \neq j.$$



Infinitesimally Rigid Deformations



Definition (Infinitesimally Rigid Deformation)

A homogeneous deformation $\phi : B \rightarrow B'$ is called *infinitesimally rigid* if its displacement field has the form

$$\mathbf{u}(\mathbf{X}) = \mathbf{c} + W\mathbf{X} \quad \text{for every } \mathbf{X} \in B,$$

where $\mathbf{c} \in \mathcal{V}$ and $W \in \mathcal{V}^2$ is skew-symmetric:

$$W^T = -W.$$

The corresponding deformation map is

$$\phi(\mathbf{X}) = \mathbf{X} + \mathbf{u}(\mathbf{X}) = \mathbf{c} + (I + W)\mathbf{X}.$$



Proposition (Infinitesimally Rigid Deformations Produce No Strain)

For an infinitesimally rigid deformation, $E(\mathbf{X}) = 0$ for every $\mathbf{X} \in B$. Thus there is no infinitesimal strain for such a deformation.

Proof.

Because $\mathbf{u}(\mathbf{X}) = \mathbf{c} + W\mathbf{X}$,

$$\nabla \mathbf{u} = W.$$

Hence

$$\begin{aligned} E &= \frac{1}{2} \left(\nabla \mathbf{u} + (\nabla \mathbf{u})^T \right) \\ &= \frac{1}{2} (W + W^T) \\ &= \frac{1}{2} (W - W) \\ &= 0. \end{aligned}$$





Geometric Consequences

Using Result 4.6, the vanishing of E implies

$$0 = E_{ii} \approx \lambda(\mathbf{e}_i) - 1 \quad (\text{no sum on } i).$$

Thus

$$\lambda(\mathbf{e}_i) \approx 1,$$

so there is no stretch to first order.

For $i \neq j$,

$$0 = E_{ij} \approx \frac{1}{2} \sin(\gamma(\mathbf{e}_i, \mathbf{e}_j)).$$

For small shear angles this gives

$$\gamma(\mathbf{e}_i, \mathbf{e}_j) \approx 0,$$

so there is no shear to first order.